

Non-VLM Card Condition Analysis Framework

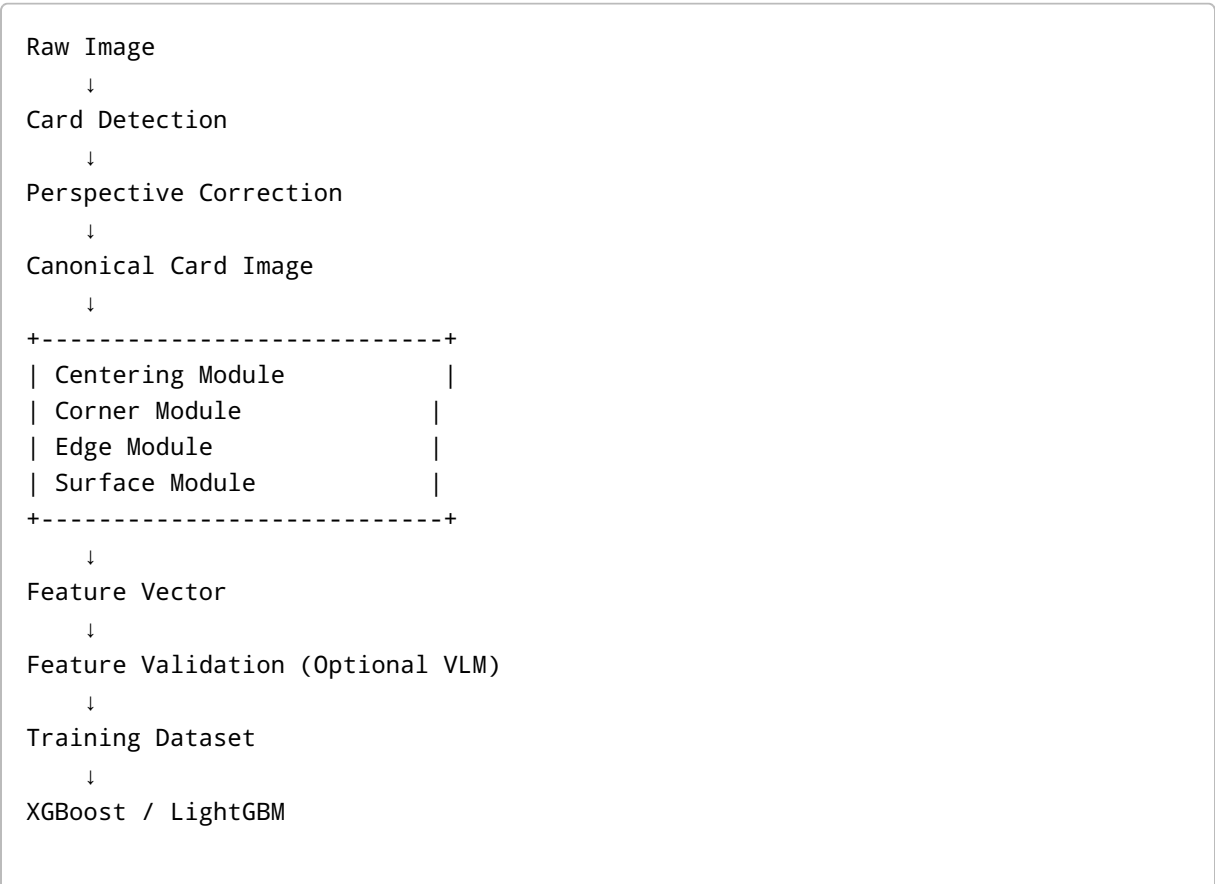
Objective

Develop a deterministic computer-vision-based grading pipeline for trading cards that minimizes dependence on VLMs.

The system should:

- 1. Compute centering deterministically.
- 2. Extract corner condition features deterministically.
- 3. Extract edge condition features deterministically.
- 4. Extract surface condition features deterministically.
- 5. Use VLMs only as secondary reviewers or feature validators.
- 6. Train a grading model using extracted features and labeled PSA grades.

High-Level Architecture



↓
PSA Grade Probabilities

Module 1: Centering

Inputs

- Warped card image
- Physical card boundary

Outputs

```
{  
  "front_lr_ratio": "55/45",  
  "front_tb_ratio": "52/48",  
  "content_region": {...}  
}
```

Approach

Step 1

Detect inner printed boundary.

Methods:

- Gradient projection
- Candidate rectangle scoring
- Border transition detection
- Segmentation refinement

Step 2

Compute:

```
left margin  
right margin  
top margin  
bottom margin
```

Step 3

Compute centering ratios.

Module 2: Corner Analysis

Goal

Measure physical wear at each corner.

Corner Crop Extraction

Extract:

```
top-left  
top-right  
bottom-right  
bottom-left
```

Corner crops should include:

- physical rounded corner
 - adjacent edges
 - surrounding border
-

Corner Features

Whitening

Definition:

```
Border color worn away exposing white cardstock.
```

Features:

```
whitening_area_ratio  
whitening_blob_count  
largest_whitening_blob
```

```
tip_whitening_present
distance_from_tip
```

Detection:

```
LAB color space
Low saturation
High lightness
Connected components
```

Fraying

Definition:

```
Visible paper fibers.
```

Features:

```
fraying_score
fiber_count
edge_jaggedness
```

Detection:

```
Contour extraction
Local roughness analysis
```

Corner Deformation

Definition:

```
Deviation from ideal rounded shape.
```

Features:

```
curve_deviation_score  
corner_radius_error  
missing_corner_area
```

Detection:

```
Fit ideal rounded corner  
Measure contour residuals
```

Corner Flattening

Definition:

```
Tip no longer follows factory radius.
```

Features:

```
tip_flattening_score  
tip_curvature
```

Corner Bending

Definition:

```
Corner not planar.
```

Single-image detection:

```
shadow asymmetry  
highlight asymmetry
```

Multi-light detection preferred.

Features:

```
bend_shadow_score  
corner_reflection_score
```

Module 3: Edge Analysis

Goal

Measure edge wear.

Edge Strip Extraction

Extract:

```
top edge  
right edge  
bottom edge  
left edge
```

Recommended width:

```
3-6% of card dimension
```

Edge Features

Whitening

Features:

```
whitening_area_ratio  
whitening_blob_count  
largest_whitening_blob
```

Detection:

LAB
HSV
Connected components

Nicks

Definition:

Small localized edge defects.

Features:

nick_count
largest_nick_depth
largest_nick_area

Chips

Definition:

Missing edge material.

Features:

chip_count
largest_chip_area
total_chip_area

Roughness

Definition:

Non-smooth edge contour.

Features:

```
roughness_score
contour_variance
edge_line_residual
```

Detection:

```
Straight-line fitting
Residual analysis
```

Module 4: Surface Analysis

Surface is the hardest component.

Recommended Capture

Single image:

```
acceptable
```

Multi-light:

```
preferred
```

Capture:

```
normal light
left raking light
right raking light
top raking light
```

Surface Features

Scratches

Features:


```
scratch_count  
largest_scratch_length  
scratch_density  
scratch_orientation_entropy
```

Detection:

```
Canny  
Sobel  
Hough lines  
Line clustering
```

Print Lines

Features:

```
print_line_count  
max_print_line_length  
print_line_contrast
```

Detection:

```
Vertical/Horizontal Sobel  
Hough transform
```

Dents

Features:

```
dent_count  
largest_dent_area  
dent_confidence
```

Best with:

```
multi-light photography
```

Creases

Features:

```
crease_count  
crease_length  
crease_area
```

Detection:

```
line detector  
shadow/highlight analysis
```

Stains

Features:

```
stain_count  
largest_stain_area  
deltaE
```

Detection:

```
LAB color anomaly detection
```

Surface Anomaly Score

Use anomaly detection.

Recommended models:

```
PatchCore  
PaDiM  
FastFlow  
Autoencoder
```

Outputs:

```
max_anomaly_score  
mean_anomaly_score  
anomaly_area_ratio
```

VLM Integration

VLM Responsibilities

VLM should NOT assign grades.

VLM should:

```
Review extracted features  
Validate ambiguous defects  
Assign artifact risk
```

Examples:

```
scratch vs foil reflection  
whitening vs border coloration  
print line vs artwork element
```

Training Dataset

Feature Vector

Centering

```
front_lr_ratio  
front_tb_ratio  
back_lr_ratio  
back_tb_ratio
```

Corners

```
whitening_area_ratio  
fraying_score  
curve_deviation_score  
tip_flattening_score  
...
```

Edges

```
whitening_area_ratio  
nick_count  
chip_count  
roughness_score  
...
```

Surface

```
scratch_count  
print_line_count  
dent_count  
crease_count  
stain_count  
anomaly_score  
...
```

VLM Validation

```
artifact_risk  
reviewer_confidence  
reviewer_agreement
```

Model Training

Recommended:

```
LightGBM
XGBoost
CatBoost
```

Targets:

```
PSA 10 probability
PSA 9 probability
PSA 8 probability
...
```

Avoid predicting only a single grade.

Prefer:

```
Probability distribution
```

Example:

```
{
  "psa10": 0.63,
  "psa9": 0.28,
  "psa8": 0.08,
  "psa7_or_lower": 0.01
}
```

Recommended Development Order

Phase 1

```
Centering
Corners
```

Phase 2

```
Edges
```

Phase 3

Surface

Phase 4

VLM feature validation

Phase 5

Train grading model

This ordering gives the fastest path to a production-quality pre-grading system.